

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |                                                                                                                                        |   |                                                                                                                                           |   |                                                                                                                                                                    |   |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------------------------------------------------------------------------------------------------------------------------------------|---|-------------------------------------------------------------------------------------------------------------------------------------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2 | 3                                                                                                                                      | 4 | 5                                                                                                                                         | 6 | 7                                                                                                                                                                  | 8 |
| A | <div>1. Connector &amp; input protection</div> <div>Field connector, 2-stage bidirectional</div> <div>TVS surge protection, chassis</div> <div>isolation</div> <div>: HMT500(260313A)_connector.kicad_sch</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   | <div>2. Power</div> <div>eFuse (reverse/OV/UV), 60V buck to 5V,</div> <div>LDO 3.3V</div> <div>: HMT500(260313A)_power.kicad_sch</div> |   | <div>3. MCU</div> <div>STM32G0B1CCT6, SWD, reset, internal</div> <div>temperature sensor</div> <div>: HMT500(260313A)_mcu.kicad_sch</div> |   | <div>4. Measurement</div> <div>Capacitive humidity sensor (PCAP04)</div> <div>and Pt1000 2-wire (ADS1220)</div> <div>: HMT500(260313A)_measurement.kicad_sch</div> |   |
| B |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |                                                                                                                                        |   |                                                                                                                                           |   |                                                                                                                                                                    |   |
| C | <div>5. Analog outputs</div> <div>2x DAC8760 V/I output with TPS26611</div> <div>miswiring protection</div> <div>: HMT500(260313A)_analog_out.kicad_sch</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   | <div>6. RS-485</div> <div>THVD2410 +/-70V fault-protected</div> <div>transceiver</div> <div>: HMT500(260313A)_rs485.kicad_sch</div>    |   | <div>7. Aux</div> <div>AO supply VAO 16.1 V (buck), supply</div> <div>monitor</div> <div>: HMT500(260313A)_aux.kicad_sch</div>            |   |                                                                                                                                                                    |   |
| D | <div>DOTECH HMT500(260313) – Oil Moisture Transmitter – Schematic v0.9 – Project No. HMT500(260313A)</div> <div>File names (schematic, PCB, Gerber) = HMT500(260313A).* PCB silkscreen marking = HMT500(260313A)</div> <div>Signal flow: J1 field connector -&gt; protection -&gt; eFuse -&gt; 5 V buck -&gt; 3.3 V LDO; sensor head J3 -&gt; PCAP04 / ADS1220 -&gt; MCU -&gt; DAC8760 x2 / THVD2410 -&gt; J1</div> <div>Outputs: RS-485 Modbus RTU + 2x analog (4-20 mA / 0-20 mA / 0-10 V / 0-5 V selectable). Supply 12-28 V DC.</div> <div>v0.10: AO D30/D40 TVS1401 +/-14 V. AO +/-28/30 V continuous miswiring is NOT guaranteed (#41).</div> <div>Power symbols: GND, +3V3, +3V3A (analog 3.3 V), +5V, VIN_P (protected input), CHASSIS. Inter-sheet signals: global labels.</div> <div>History v0.3-v0.8 (MCU G0B1, JST harness, DAC8760/TPS26611 datasheet pinouts, RTN isolation, gated DAC SCLK ...): docs/decision-log.md #9-#26.</div> <div>v0.9: 3-pass review (docs/hw/final-review-260313A.md): OVP 32.6 V (R5 36.5k); AO supply VAO 16.1 V buck U15 (DAC AVDD, TPS26611 +Vs; dec. #37, IVS320 AO); D3 series Schottky; RS-485 TVS D60/D61; D2 -&gt; TVS3301; eFuse IN C2; buck CIN:</div> <div>RS485_DE 10k pull-down (in-house bootloader, dec. #31); THVD2410; X7R only; R2 HV 1206; SGOOD -&gt; PA0/PA4; CS_CDC -&gt; PA15; VIN_SENSE PB0;</div> <div>LATCH/CS pull-ups; OPA197 V+ = DAC AVDD; R31/R41 100k; SWD no-connect flags removed; sheet Aux: +Vs clamp, VIN monitor (DAC boost / HSE / sensor ESD: no room, dec. #32).</div> <div>Footprints: KiCad 7.0.11 library + project library HMT500_260313A (GDT Bourns 2035-xx-SM, TVS3301 pad floating).</div> <div>표기 규칙: 부품번호 = 굵은 글자 (R1, U4) / 부품값 = 보통 글자 (10k, DAC8760) / 전원 네트 = 가울인 글자 (+3V3, VIN_P)</div> <div>신호 네트 = 테두리 있는 라벨 (SPL_SCK, OUT1_EXT) - 네트 이름은 부품번호-부품명과 겹치지 않게 지음</div> <div>라벨 모양 = 신호 방향: 뾰족한 쪽이 밖 = 이 시트에서 내보냄(출력), 안 = 받음(입력), 양쪽 = 양방향, 네모 = 아날로그-수동</div> <div>글자 겹침: 생성기가 부품 몸체-핀-선-라벨-메모와 겹치지 않는 자리에 자동 배치하고, 겹치면 생성 실패로 처리</div> |   |                                                                                                                                        |   |                                                                                                                                           |   |                                                                                                                                                                    |   |
| E |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |                                                                                                                                        |   |                                                                                                                                           |   |                                                                                                                                                                    |   |
| F | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2 | 3                                                                                                                                      | 4 | 5                                                                                                                                         | 6 | 7                                                                                                                                                                  | 8 |

Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)

Design notes: docs/hw/circuit-design.md

Project No. HMT500(260313A) – HMT500(260313) oil moisture transmitter

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Sheet: /

File: HMT500(260313A).kicad\_sch

Title: HMT500(260313A) Top

Size: A3

Date: 2026-10-01

Rev: 0.10

KiCad E.D.A. 10.0.6

Id: 1/8

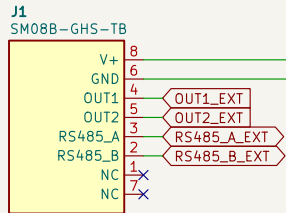
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| Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib) |                  |           |
| Design notes: docs/hw/circuit-design.md                                  |                  |           |
| Project No. HMT500(260313A) – HMT500(260313) oil moisture transmitter    |                  |           |
| DOTECH Co., Ltd.                                                         |                  |           |
| Sheet: /                                                                 |                  |           |
| File: HMT500(260313A).kicad_sch                                          |                  |           |
| Title: HMT500(260313A) Top                                               |                  |           |
| Size: A3                                                                 | Date: 2026-10-01 | Rev: 0.10 |
| KiCad E.D.A. 10.0.6                                                      |                  | Id: 1/8   |

### FIELD CONNECTOR

현장 커넥터

J1 JST GH 8P -> W-2 -> M12 8P

EE364 pinout, pin n = pin n



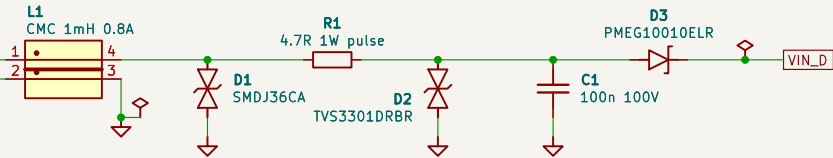
Pins 1, 7: not connected

GH mounting pads (MP): PCB only

### INPUT SURGE / REVERSE-POLARITY PROTECTION

입력 서지·역극성 보호

TVS bidirectional: -30 V miswiring must not conduct



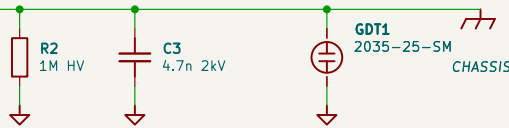
SMDJ (1st) -> R1 -> TVS3301 (2nd) -> D3 100 V Schottky -> eFuse (sheet Power)

J5  
CHASSIS wire

CHASSIS

### CIRCUIT GND <--> CHASSIS (floating)

회로 GND-외함 분리



PCB-housing creepage >= 2 mm

GDT conducts only on line-to-ground surge

### 주요 부품 — 역할 · 특징 · 사양

| 부품번호   | 부품               | 역할              | 특징 · 사양                                                                                                                                 |
|--------|------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| J1     | SM08B-GHS-TB     | 현장 커넥터 연결       | JST GH 1.25 mm 8P 옆 삽입, 하네스 W-2 → M12 8P, 핀 n = M12 핀 n                                                                                 |
| L1     | CMC 1mH 0.8A     | 전원선 공통모드 노이즈 차단 | Bourns SRF0905-102Y (LCSC, 744222 동등), 2 × 1 mH, 0.8 A, 9.2 × 6 × 5.3 mm — 전도 방출·내성 대책                                                  |
| D1     | SMDJ36CA         | 입력 서지 1단 흡수     | TVS 양방향 3000 W, 36 V — -30 V 오결선에 도통 안 함                                                                                                |
| R1     | 4.7R 1W pulse    | 1단·2단 서지 분담     | 2512 1 W (Vishay CRCW2512, JLC 부품 — 결정 #38, 펄스 에너지 시험 T1)                                                                               |
| D2     | TVS3301DRBR      | 입력 서지 2단 클램프    | TI 평탄 클램프 ±33 V, 최대 42.5 V @ 27 A, DC 입력 33 V 초과 금지 (v0.9, 이전 SMBJ33CA)                                                                 |
| D3     | PMEG10010ELR     | 음(-) 서지·역극성 차단  | Nexperia 100 V 1 A 쇼트키 (저누설). D2가 -42.5 V로 갇는 동안 출력 쪽 +28 V → 역전압 약 71 V를 D3가 받음 → eFuse IN-OUT 역전압 없음 (외부 검수 H01). 손실 약 0.45 V × 0.1 A |
| GDT1   | 2035-25-SM       | 회로 GND-외함 서지 방전 | Bourns 2전극 SMD GDT 250 V (LCSC), Ø5 × 4.4 mm — 선-대지 서지 때만 도통                                                                            |
| C3, R2 | 4.7n 2kV / 1M HV | GND-외함 고주파 결합   | 4.7 nF 2 kV X7R 1812 + 1 MΩ 서지용 칩 저항 1206 (500 V, GDT 방전 전 임펄스, 물딩 안)                                                                   |
| J5     | CHASSIS wire     | PCB-바다 접지 선     | AWG 28 선 막 25 mm → M2 링 단자 → PCB 홀더 축 나사(금속 바다 탭). 조립 시뮬레이션 결과 스프링 접촉 대체                                                                |

\* 표시 값은 데이터시트로 확인

Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)  
Design notes: docs/hw/circuit-design.md  
Project No. HMT500(260313A) — HMT500(260313) oil moisture transmitter

**DOTECH Co., Ltd.**

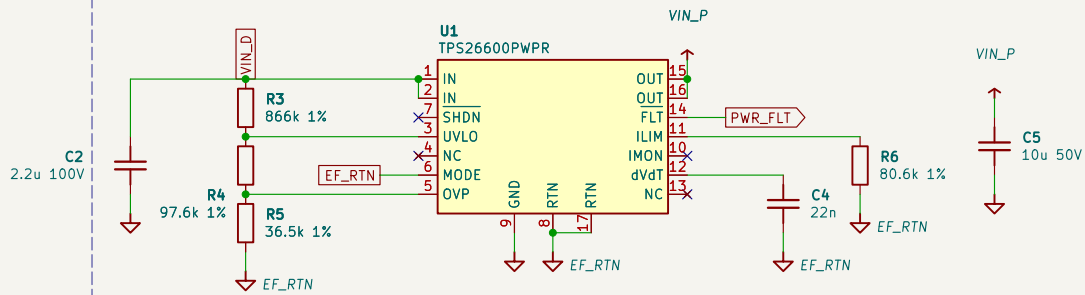
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File: HMT500(260313A)\_connector.kicad\_sch

**Title: HMT500(260313A) Connector & input protection**

|                     |                  |           |
|---------------------|------------------|-----------|
| Size: A3            | Date: 2026-10-01 | Rev: 0.10 |
| KiCad E.D.A. 10.0.6 |                  | Id: 2/8   |

eFuse TPS2660 reverse -60 V / OVP 32.6 V / UVLO 8.9 V

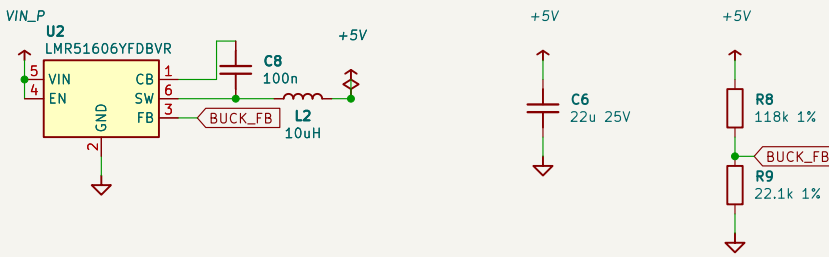
전자 퓨즈



$UVLO = 1.19 \text{ V} \times (R3+R4+R5)/(R4+R5) = 8.9 \text{ V}$      $OVP \text{ rise } 1.19 \text{ V} \times (R3+R4+R5)/R5 = 32.6 \text{ V} (31.5-34.2), \text{ fall } 30.1 \text{ V}$   
 $L\_OL = 12 / R6 = 149 \text{ mA}$ . Supply 12-28 V recovers after OVP trip (fall > 29.2 V). TPS26611 +Vs clamped (sheet Aux).  
EF\_RTN = TPS2660 RTN reference: R5, R6, C4, MODE, PowerPAD. NEVER connect to GND (datasheet 9.3.5.5).

BUCK 5 V LMR51606 (4-65 V in, 0.6 A, 1.1 MHz)

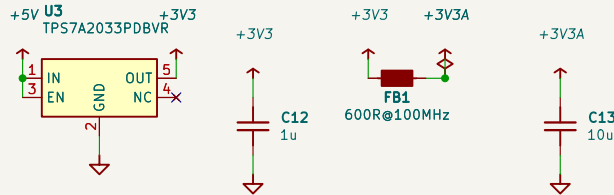
5 V 강압 전원



$V_{out} = 0.8 \text{ V} \times (1 + R8/R9) = 5.07 \text{ V}$ . 1.1 MHz: L 10 uH, COUT 22 uF (SLUSEY1B Table 8-1). Same IC as U15.

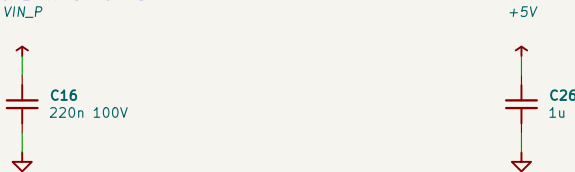
LDO 3.3 V TPS7A2033 + analog rail +3V3A

3.3 V 레귤레이터



LOCAL INPUT CAPS

국부 입력 콘덴서



C16 at U2 VIN-GND, C5 (bulk) next to U2; C26 at U3 IN

주요 부품 — 역할 · 특징 · 사양

| 부품번호      | 부품             | 역할               | 특징 · 사양                                                                                                                             |
|-----------|----------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| U1        | TPS26600PWPR   | 전자 퓨즈 (입력 보호)    | 역극성 -60 V 차단, 과전압 32.6 V (복귀 30.1 V)-저전압 8.9 V (R3-R5), 전류 제한 149 mA (R6), 돌입 제한 (C4), 고장 출력 FLT                                    |
| C2        | 2.2u 100V      | eFuse 입력 콘덴서     | TVS와 함께 $IN \geq 1 \text{ uF}$ (데이터시트 11.1), U1 핀 옆 — 핫플러그 링잉 억제                                                                    |
| EF_RTN    | (기준 접지)        | TPS2660 내부 기준    | R5-R6-C4-MODE-방열 패드는 RTN에 연결. GND와 연결 금지 — 역극성 보호 무효-손상                                                                             |
| U2        | LMR51606YFDBVR | 5 V 강압 전원 (벅)    | 입력 4-65 V, 0.6 A 동기식, 1.1 MHz FPWM, SOT-23-6, 출력 = $0.8 \text{ V} \times (1 + R8/R9) = 5.07 \text{ V}$ (결정 #38, JLC 부품, U15와 같은 IC) |
| L2, C6    | 10uH / 22u 25V | 벅 출력 필터          | SLUSEY1B 표 8-1 (5 V, 1.1 MHz: 10 uH, $\geq 10 \text{ uF}$ ), L2 SMNR4020-10UH 4 × 4 × 2 mm                                          |
| C16 (+C5) | 220n (+10u)    | 벅 입력 콘덴서         | VIN-GND 핀 옆 220 nF, 벌크 C5 10 uF를 U2 옆에                                                                                              |
| U3        | TPS7A2033PDBVR | 3.3 V 저잡음 레귤레이터  | 300 mA, 저잡음-높은 PSRR → 벅 리플 제거, SOT-23-5                                                                                             |
| FB1       | 600R@100MHz    | 아날로그 전원 +3V3A 분리 | 페라이트 비드 + C13 10 uF → PCAP04-ADS1220 전원                                                                                             |

• 표시 값은 데이터시트로 확인

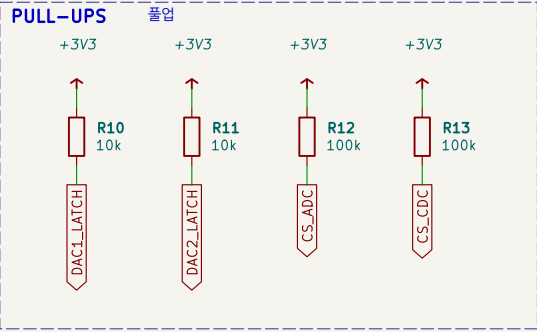
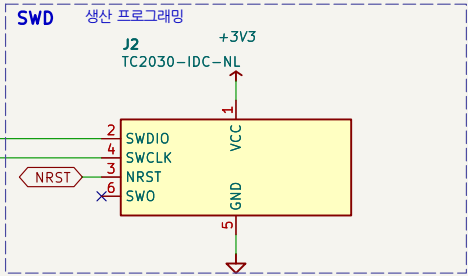
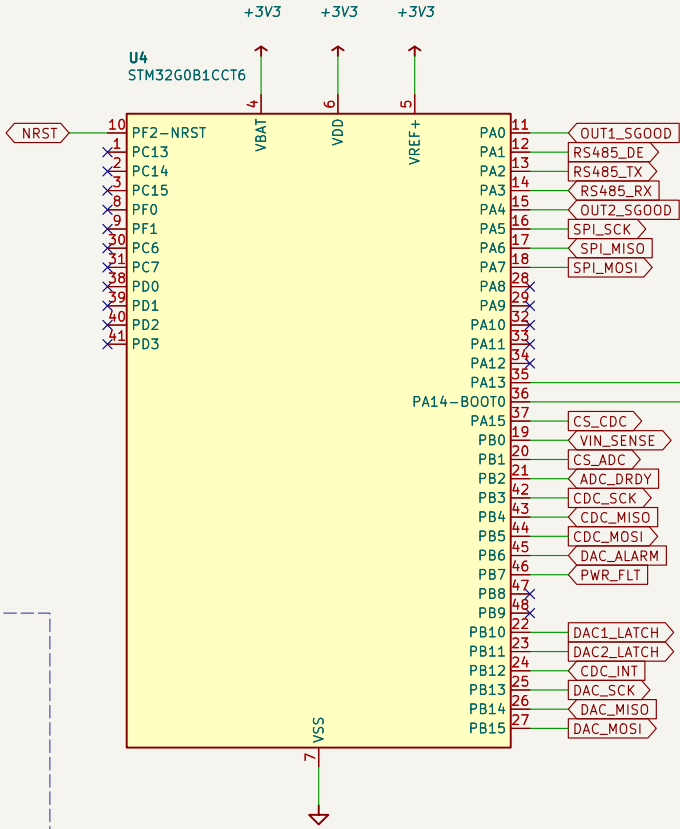
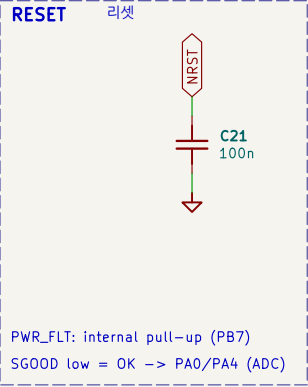
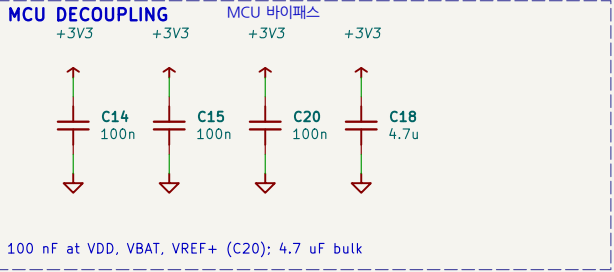
Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)  
Design notes: docs/hw/circuit-design.md  
Project No. HMT500(260313A) — HMT500(260313) oil moisture transmitter

DOTECH Co., Ltd.

Sheet: /2. Power/  
File: HMT500(260313A)\_power.kicad\_sch

Title: HMT500(260313A) Power

|                     |                  |           |
|---------------------|------------------|-----------|
| Size: A3            | Date: 2026-10-01 | Rev: 0.10 |
| KiCad E.D.A. 10.0.6 |                  | Id: 3/8   |



**MCU NOTES (STM32G0B1)** MCU 설정 메모

BOOT0 shares PA14/SWCLK - no pull-down.

Option bytes: nBOOT\_SEL=1, nBOOT0=1, IWDG\_SW=0, RDP 1

-> in-house RS-485 bootloader 0x08000000 (dec. #31).

VDDA = VDD (LQFP48); VREF+ tied to +3V3 (C20).

PCB temp (mid): internal sensor, ADC ch TS + TS\_CAL.

Cortex-M0+: no SWO (J2 pin 6 NC).

Same die as DP2000 (G0B1CCT6); T3 = -40..125 C.

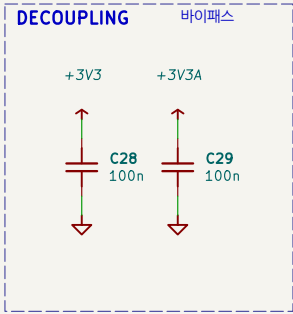
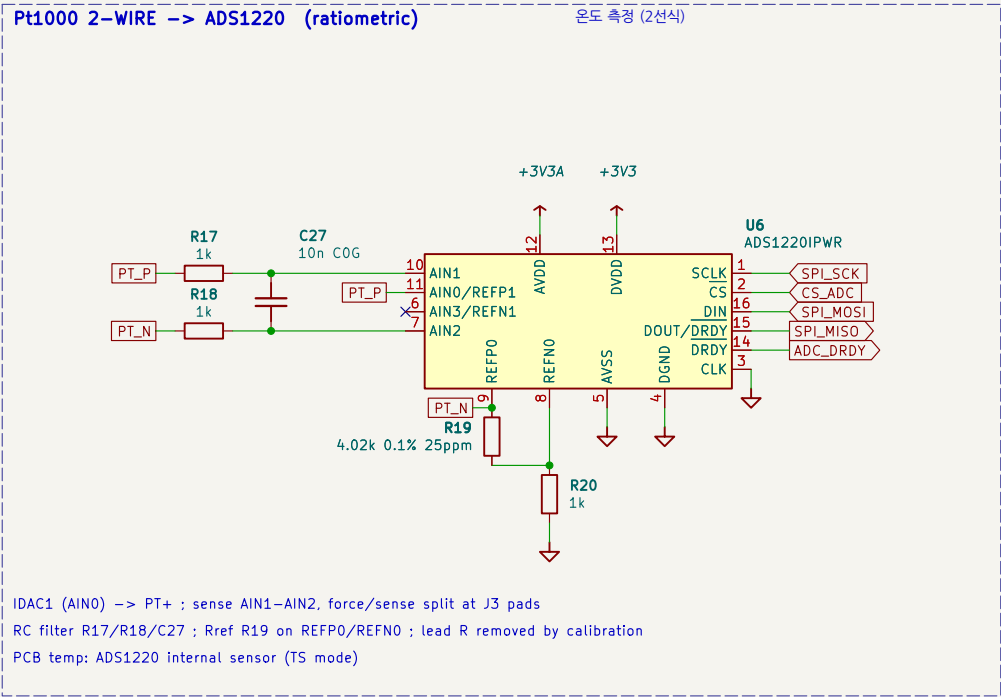
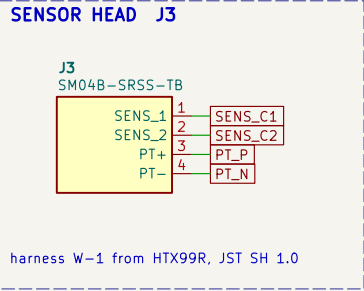
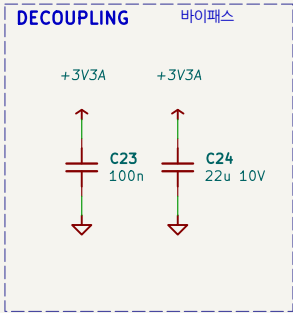
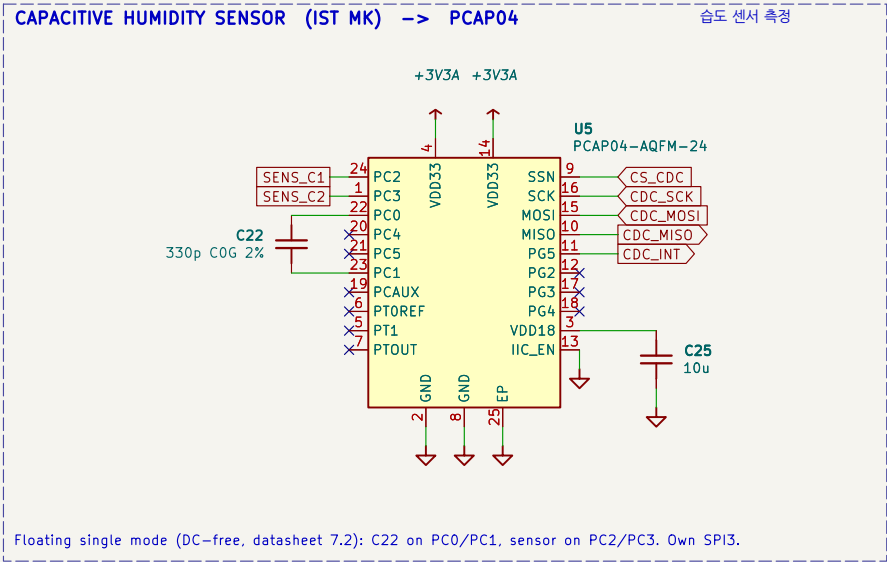
VIN\_SENSE (PB0 ADC\_IN8) = VIN\_P / 11.

| 주요 부품 — 역할 · 특징 · 사양 |               |                   |                                                                                                                                                     |
|----------------------|---------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 부품번호                 | 부품            | 역할                | 특징 · 사양                                                                                                                                             |
| U4                   | STM32G0B1CCT6 | 계어통신·보정 연산        | Cortex-M0+ 64 MHz, 플래시 256 KB, RAM 144 KB, -40-85 °C (샘플 #39), LQFP48, 내부 온도센서 (PCB 온도), SPI1 = ADS1220, SPI2 = DAC (모드 3, SCLK 게이트), SPI3 = PCAP04 |
| J2                   | TC2030-IDC-NL | 생산 때 펌웨어 기록 (SWD) | PCB 패드만 (부품 없음). 이후 업데이트는 RS-485 부트로더                                                                                                               |
| C21                  | 100n          | 리셋 노이즈 필터         | NRST 핀, 물딩 후 EMC 여유                                                                                                                                 |
| R10, R11             | 10k           | DAC LATCH 풀업      | 리셋 중 LATCH High -> SCLK 게이트 닫힘, DAC 오입력 방지 (v0.9)                                                                                                   |
| R12, R13             | 100k          | CS 풀업             | 리셋 중 ADS1220-PCAP04 선택 해제 (디지털 입력 부동 금지)                                                                                                            |

\* 표시 값은 데이터시트로 확인

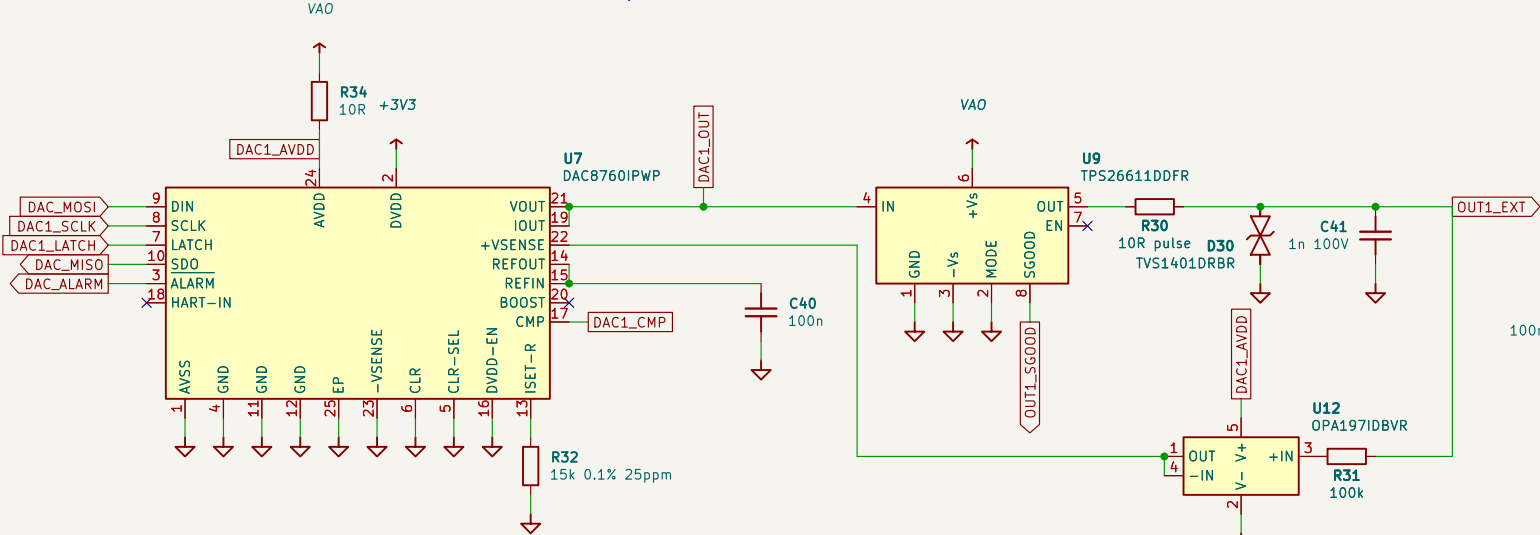
Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)  
Design notes: docs/hw/circuit-design.md  
Project No. HMT500(260313A) - HMT500(260313) oil moisture transmitter  
**DOTECH Co., Ltd.**

|                                                        |                  |           |
|--------------------------------------------------------|------------------|-----------|
| Sheet: /3. MCU/<br>File: HMT500(260313A)_mcu.kicad_sch |                  |           |
| <b>Title: HMT500(260313A) MCU</b>                      |                  |           |
| Size: A3                                               | Date: 2026-10-01 | Rev: 0.10 |
| KiCad E.D.A. 10.0.6                                    |                  | Id: 4/8   |



| 주요 부품 — 역할 · 특징 · 사양 |                  |                   |                                                                                                                   |
|----------------------|------------------|-------------------|-------------------------------------------------------------------------------------------------------------------|
| 부품번호                 | 부품               | 역할                | 특징 · 사양                                                                                                           |
| J3                   | SM04B-SRSS-TB    | 센서 하네스 W-1 연결     | JST SH 1.0 mm 4P 옆 삽입, 1:2 = MK33, 3:4 = Pt1000                                                                   |
| U5                   | PCAP04-AQFM-24   | 습도 센서 정전용량 측정     | 정전용량-디지털 변환, 기준 C와 비율 측정(방전 시간), DC 없는 구동, 내부 DSP, 전용 SPI3 (모드 1)                                                 |
| C25, C24             | 10u / 22u        | PCAP04 전원 버퍼      | VDD18 ≥ 4.7 uF, VDD33 ≥ 10 uF (데이터시트 표 2) — DC 바이어스 후에도 총축 (v0.9)                                                 |
| C22                  | 330p COG 2%      | PCAP04 기준 용량      | 센서 MK33-W (300 pF, 결정 #33) 측정 범위 가운데 값, COG                                                                       |
| U6                   | ADS1220IPWR      | Pt1000 온도 측정      | 24비트 ADC, 50/60 Hz 제거, 내부 온도센서 (PCB 온도), 설정: IDAC = 250 uA 고정 (100 uA면 기준 0.40 V < 권장 0.75 V), PGA ≤ 2 (SP1 모드 1) |
| R19                  | 4.02k 0.1% 25ppm | Pt1000 비율 측정 기준저항 | IDAC 전류 오차 상쇄, 온도계수 25 ppm/°C (결정 #38)                                                                            |
| R17, R18, C27        | 1k / 10n COG     | 입력 RC 필터          | 차동 노이즈 제거 (COG, 데이터시트 9.2)                                                                                        |
| * 표시 같은 데이터시트로 확인    |                  |                   |                                                                                                                   |

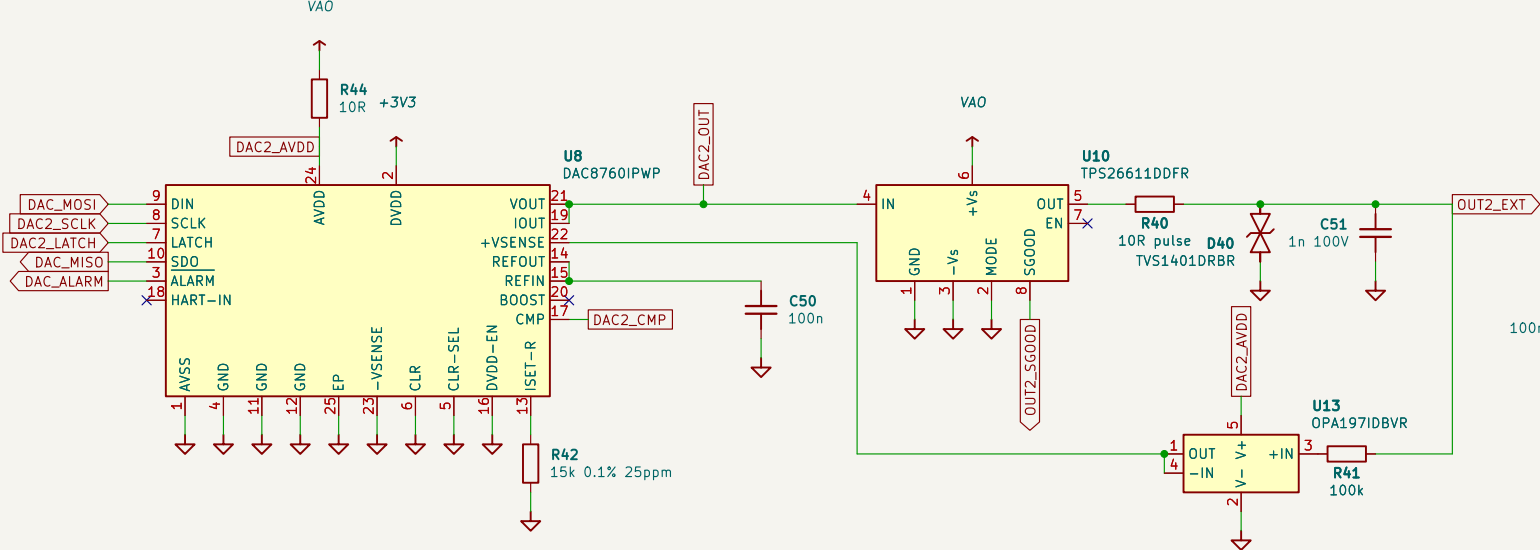
## ANALOG OUTPUT CH1 - V/I selectable, TVS +/-14 V -&gt; connector pin 4



VOUT + IOOUT one terminal (datasheet 9.1.1.3). +VSENSE = U12 OPA197 follower of terminal OUT1\_EXT (after R30) via R31:  
no +VSENSE 60k leakage in current mode, series-R drop corrected in voltage mode. ISET-R = R32 15k 0.1% 25ppm. SGOOD low = OK.

## 아날로그 출력 1

## ANALOG OUTPUT CH2 - V/I selectable, TVS +/-14 V -&gt; connector pin 5

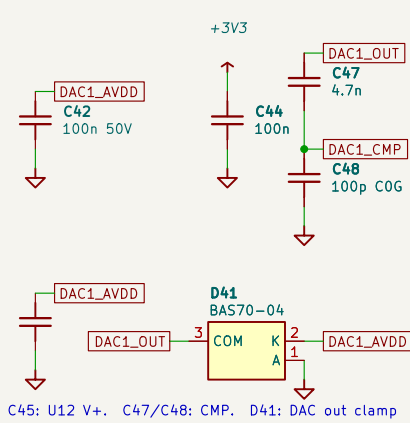


VOUT + IOOUT one terminal (datasheet 9.1.1.3). +VSENSE = U13 OPA197 follower of terminal OUT2\_EXT (after R40) via R41:  
no +VSENSE 60k leakage in current mode, series-R drop corrected in voltage mode. ISET-R = R42 15k 0.1% 25ppm. SGOOD low = OK.

## 아날로그 출력 2

## CH1 DECOUPLING

## 채널 1 바이패스



C45: U12 V+. C47/C48: CMP. D41: DAC out clamp

## DAC SELECTION (decision #10)

## DAC 선정

U7/U8 fitted: DAC8760 (16-bit, I + V output).

Drop-in alternates (same pinout, no layout change):

DAC7760 12-bit, I + V (cost-down, V/I kept)

DAC8750 16-bit, I only 4-20/0-20/0-24 mA (EE364-equiv.)

DAC7750 12-bit, I only (-5 uA/step vs EE364 2 uA)

Current-only (x750): VOOUT/+VSENSE/-VSENSE absent -

check pin handling before fitting (VERIFY).

## DAC SPI - SPI2, SCLK gated per DAC

## DAC 전용 SPI

DAC8760 rev D: daisy chain removed; multiple devices

need gated SCLK (datasheet 8.5.1.5, Fig. 8-9).

SPI2 mode 3 (CPOL 1, CPHA 1): SCK idles high, so the

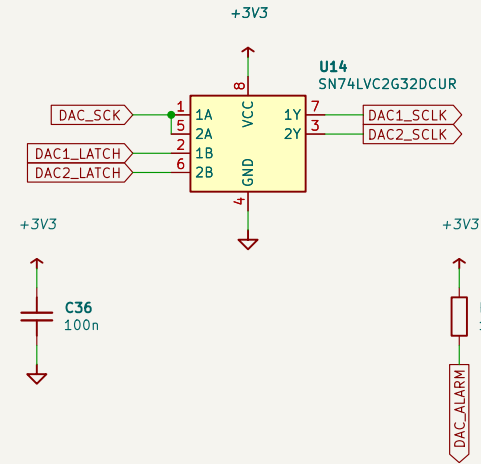
OR gate gives no extra edge at LATCH low/high.

DACn\_SCLK = DAC\_SCK OR DACn\_LATCH (PB10 / PB11).

DIN and SDO shared (SDO 3-state after LATCH).

ALARM: open drain, wired-OR, R33 10k pull-up:

which channel -> read status register.



## 주요 부품 - 역할 · 특징 · 사양

| 부품번호             | 부품               | 역할                | 특징 · 사양                                                                                                                  |
|------------------|------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------|
| U7, U8           | DAC8760IPWP      | 아날로그 출력 1·2       | 16비트, 4-20 / 0-20 / 0-24 mA · 0-5 / 0-10 V 선택, 내부 기준, 고장 알람 — 대체 DAC7760/8750/7750                                       |
| U9, U10          | TPS26611DDFR     | 출력 오걸선 보호         | ±50 V, 단자 > +Vs 또는 < -0.2 V면 5 us 차단·자동 복귀, 전류 제한 32 mA, RON 7.5 Ω, SGOOD Low = 정상. +Vs = VAO 16.1 V (백, 시트 Aux, 결정 #37) |
| R30, R40         | 10R pulse (2512) | 서지 전류 제한          | 출력 보호기와 TVS 사이 직렬                                                                                                        |
| D30, D40         | TVS1401DRBR      | 출력 서지 클램프         | ±14 V 동작, 클램프 최대 23.55 V @ 30 A/125 C, AO ±28/30 V 지속 오걸선 보장 제외 (#41)                                                    |
| C41, C51         | 1n 100V          | 출력 고주파 필터         | 케이블로 들어오는 RF 억제                                                                                                          |
| U12, U13         | OPA197IDBVR      | +VSENSE 버퍼 (이득 1) | 36 V, 입출력 레일투레일 — 단자 전압을 +VSENSE로, 전류 모드 누설 0. V+ = DAC AVDD                                                             |
| R31, R41         | 100k             | 버퍼 입력 보호          | R30-R40 뒤 단자 전압을 OPA197 + 입력으로 (서지 전류 제한, ±30 V 오배선 9 mW, v0.9)                                                          |
| R32, R42         | 15k 0.1% 25ppm   | 외부 ISET-R (전류 기준) | 내부 저항보다 온도 드리프트 작음 (REXT 비트 설정)                                                                                          |
| R34, R44         | 10R              | DAC AVDD 직렬       | VAO → AVDD, 전원 상승 속도 제한 (데이터시트 10장)                                                                                      |
| C47/C48, C57/C58 | 4.7n / 100p C0G  | DAC CMP 보상        | VOOUT-CMP 4.7 nF + CMP-GND 100 pF (SBA5528D 8.3.2: 470 pF 초과 시 100 pF 추가) — 케이블 용량·보호기 RON 포함 루프 안정 (IVS320 AO rev 1.0)  |
| D41, D51         | BAS70-04         | DAC 출력 클램프        | DAC 출력 줄을 GND - 0.3 V ~ AVDD + 0.3 V로 — TPS26611 차단-재연결 순간 DAC 핀 보호 (외부 검수 H03, IVS320과 동일)                              |
| U14              | SN74LVC2G32DCUR  | DAC별 SCLK 게이트     | OR 2회로: DACn_SCLK = SCK OR LATCHn (SPI2 모드 3)                                                                            |
| R33              | 10k              | ALARM 플립          | 오픈 드레인 ALARM 두 개를 한 선으로 (데이터시트: 외부 10 kΩ)                                                                                |

\* 표시 같은 데이터시트로 확인

Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)

Design notes: docs/hw/circuit-design.md

Project No. HMT500(260313A) — HMT500(260313) oil moisture transmitter

DOTEC Co., Ltd.

Sheet: /5. Analog outputs/

File: HMT500(260313A)\_analog\_out.kicad\_sch

Title: HMT500(260313A) Analog outputs

Size: A3

Date: 2026-10-01

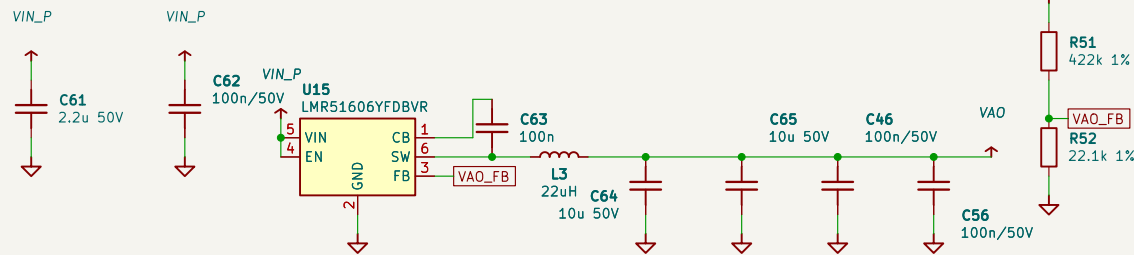
Rev: 0.10

KiCad E.D.A. 10.0.6

Id: 6/8



AO SUPPLY VAO 16.1 V - DC-DC (decision #37)



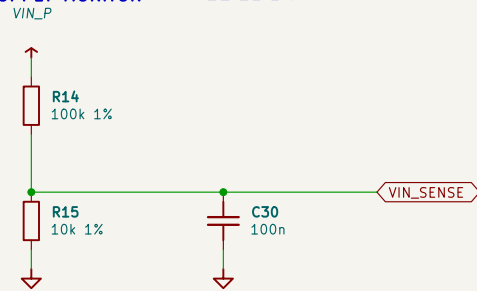
VAO = 0.8 V x (1 + 422k/22.1k) = 16.08 V (15.5–16.6 V). LMR51606 FPWM 1.1 MHz, max duty 98 %.

Load 2 x (DAC 27 + OPA197 1.3 + TPS26611 1.75 mA) = 60 mA. C46/C56 at U9/U10 pin 6.

VIN\_P < ~17 V: VAO follows VIN\_P (dropout). 24 mA x 500 ohm OK down to VIN ~15.8 V.

## SUPPLY MONITOR

전원 전압 감시



VIN\_SENSE = VIN\_P / 11 -> PB0 (ADC\_IN8)

OPTIONS NOT FITTED ON PCB

PCB에 넣지 않은 선택 사항

DAC8760 IOUT boost: not needed – AVDD = VAO 16.1 V (decision #37) cuts DAC heat

from about 0.66 W to 0.39 W per channel (24 mA short, any supply 12–28 V).

Thermal test on a potted prototype still planned.

HSE crystal: no room next to PF0/PF1 -> factory HSI trim + firmware HSI calibration

from Modbus frame timing (USART auto-baud / timer capture).

Sensor-line ESD array: no room next to J3 -> probe ESD test (IEC 61000-4-2) decides;

if needed: protect at the HTX99R side or in the next PCB rev.

## 주요 부품 — 역할 · 특징 · 사양

| 부품번호                 | 부품                     | 역할                 | 특징 - 사양                                                                                                         |
|----------------------|------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>U15</b>           | LMR51606YFDBVR         | AO 전원 VAO 16.1 V 벅 | 4-65 V 입력 0.6 A, 1.1 MHz FPWM, SOT-23-6 — DAC AVDD-TPS26611 +Vs-OPA197 V+ 공용 (IVS320 AO rev 1.0과 같은 회로, 결정 #37) |
| <b>L3, C64, C65</b>  | 22uH / 10u 50V x2      | 벅 출력               | 22 uH 1.05 A 4 × 4 × 2 mm, 10 uF 50 V X7R 1206 × 2 (16 V에서 실효 약 4 uF × 2)                                       |
| <b>R51, R52</b>      | 422k / 22.1k 1%        | VAO 설정             | $0.8 \text{ V} \times (1 + 422/22.1) = 16.08 \text{ V}$                                                         |
| <b>C61, C62, C63</b> | 2.2u 50V / 100n / 100n | 입력-부트스트랩           | VIN 핀 옆 입력 C, CB-SW 100 nF                                                                                      |
| <b>C46, C56</b>      | 100n/50V               | +Vs 바이패스           | C46 → U9 6번, C56 → U10 6번 핀 옆                                                                                   |
| <b>R14, R15, C30</b> | 100k / 10k / 100n      | 전원 전압 감시           | VIN_P / 11 → PB0 ADC (UVLO 8.9 V ~ DAC 최소 10 V 사이 판정)                                                           |

\* 표시 값은 데이터시트로 확인

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Project No. HMT500(260313A) - HMT500(260313) oil moisture transmitter

**DO TECH Co., Ltd.**

Sheet: /7. Aux/  
File: HMT500(260313A)\_aux.kicad\_sch

Title: HMT500(260313A) Aux

|                     |                  |           |
|---------------------|------------------|-----------|
| Size: A3            | Date: 2026-10-01 | Rev: 0.10 |
| KiCad E.D.A. 10.0.6 |                  | Id: 8/8   |